

Homework 3
MSBA 400: Statistical Foundations for Data Analytics
Professor Rossi

Question 1 : Prediction from Multiple Regressions

Q1, part A

Run the multiple regression of `Sales` on `p1` and `p2` using the dataset, `multi`.

Q1, part B

Suppose we wish to use the regression from part A to estimate sales of this firm's product with, `p1` = \$7.5. To make predictions from the multiple regression, we will have to predict what `p2` will be given that `p1` = \$7.5.

Explain why setting `p2=mean(p2)` would be a bad choice. Be specific and comment on why this is true for this particular case (value of 'p1').

Q1, part C

Use a regression of `p2` on `p1` to predict what `p2` would be given that `p1` = \$7.5.

Q1, part D

Use the predicted value of `p2` from part C, to predict `Sales`. Show that this is the same predicted value of sales as you would get from the simple regression of `Sales` on `p1`. Explain why this must be true.

Question 2: Interactions

An interaction term in a regression is formed by taking the product of two independent or predictor variables as in:

$$Y_i = \beta_0 + \beta_1 X1_i + \beta_2 X2_i + \beta_3 X1_i * X2_i + \varepsilon_i$$

This term has a non-linear effect, which allows the effect of variable `X1` to be moderated by the level of `X2`. We can take the partial derivative of the conditional mean function to see this:

$$\frac{\partial}{\partial X1} E[Y|X1, X2] = \beta_1 + \beta_3 X2$$

Return to the regression in Chapter 6 of `log(emv)` on `luxury`, `sporty` and add the interaction term `luxury*sporty`.

Q2, part A

Compute the change in `emv` we would expect to see if `sporty` increased by .1 units, holding `luxury` constant at .30 units

Q2, part B

Compute the change in `emv` we would expect to see if `sporty` was increased by .1 units, holding `luxury` constant at .70 units.

Q2, part C

Why are the answers different in part A and part B? Does the interaction term make intuitive sense to you? Why?

Question 3: More on ggplot2 and regression planes

The classic dataset, `diamonds`, (you must load the `ggplot2` package to access this data) has about 50,000 prices of diamonds along with weight (`carat`) and quality of cut (`cut`).

1. Use `ggplot2` to visualize the relationship between price and carat and cut. ‘price’ is the dependent variable. Consider both the `log()` and `sqrt()` transformation of price.
2. Run a regression of your preferred specification. Perform residual diagnostics. What do you conclude from your regression diagnostic plots of residuals vs. fitted and residuals vs. carat?

note: `cut` is a special type of variable called an ordered factor in R. For ease of interpretation, convert the ordered factor into a “regular” or non-ordinal factor.

```
library(ggplot2)
data(diamonds)
cutf=as.character(diamonds$cut)
cutf=as.factor(cutf)
```